Raman Spectroscopy

Parth Bhargava (A0310667E)
Experimental Physics II
Experiment B

Raman spectroscopy with a 532 nm laser was used to characterise carbon allotropes, assess three diamond-like samples, estimate the thickness of a graphene film, and identify an unlabeled semiconductor wafer. Graphite (carbon rod), HOPG, carbon nanotubes, and diamond each produced distinct spectra consistent with their bonding geometry, with measured D, G, and 2D band positions falling within standard literature ranges. Background-subtracted D/G intensity ratios ranged from 0.04 ± 0.01 (HOPG) to 1.2 ± 0.3 (nanotubes), and G-band linewidths (FWHM 18–65 cm^{-1}) tracked the same disorder trend. Among the three diamond-like samples, one (DLC-I) yielded a resolvable Raman spectrum with a D band at 1300 ± 5 cm^{-1} and a G band at 1559 ± 5 cm^{-1} (FWHM $\approx$ 180–200 cm^{-1}, $\frac{I_D}{I_G} = 0.87$), consistent with amorphous carbon (a-C); the other two were dominated by photoluminescence characteristic of hydrogen-rich a-C:H. None of the three produced the sharp 1332 cm^{-1} diamond peak. The graphene spectrum showed a 2D/G intensity ratio of 0.87 with a 2D FWHM of 54 cm^{-1}, placing it in the few-layer regime. The unknown wafer was identified as crystalline silicon from its characteristic mode at 521 cm^{-1}.

1. INTRODUCTION

Most analytical techniques require at least some sample preparation, but Raman spectroscopy is unusual in that it works on materials essentially as they are. A laser beam focused onto the sample scatters off the lattice vibrations, and the energy shift of the scattered photons encodes the vibrational frequencies of the material. Because different crystal structures, bonding geometries, and even the number of atomic layers all affect these frequencies, a single Raman spectrum can reveal a great deal about a sample's identity and structural quality.

This is particularly useful for carbon, which exists in several allotropic forms. Graphite, diamond, nanotubes, and graphene all contain only carbon atoms but differ fundamentally in their bonding. sp^2 carbon (graphite, graphene, nanotubes) and sp^3 carbon (diamond) produce entirely different Raman spectra, and within the sp^2 family, disorder, stacking order, and dimensionality introduce further spectral differences. Raman spectroscopy exploits exactly these differences to distinguish materials that would be chemically identical by elemental analysis.

This experiment applies Raman spectroscopy to four problems: identifying the characteristic vibrational bands of several carbon allotropes, determining whether three unknown samples are genuine diamond, estimating the layer count of a graphene film on a SiO_2/Si substrate, and identifying an unlabeled semiconductor wafer.

2. THEORY

2.1 Raman Scattering

When monochromatic laser light hits a material, most photons scatter elastically, leaving with the same energy they arrived with. This is Rayleigh scattering. A small fraction of photons, however, exchange energy with lattice vibrations (phonons) and scatter inelastically. If a photon gives energy to the lattice, it leaves with less energy and a longer wavelength; this is Stokes scattering. If instead it picks up energy from a phonon that was already thermally excited, it leaves with more energy; this is anti-Stokes scattering.

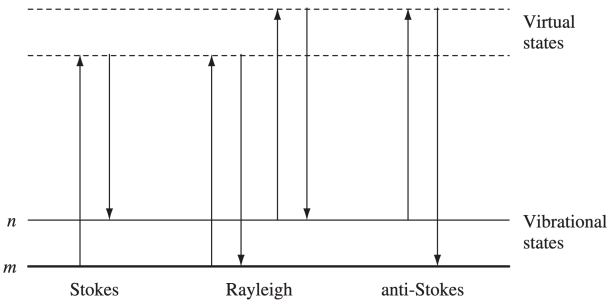


FIG. 1. Rayleigh, Stokes, and anti-Stokes scattering energy-level diagram. The Stokes process leaves the molecule in a higher vibrational state; the anti-Stokes process starts from an excited state and returns the molecule to the ground state [1].

The Raman spectrum plots scattered intensity against the frequency shift relative to the laser line, measured in wavenumber units:

$$\Delta\tilde{\nu} = \frac{1}{\lambda_0} - \frac{1}{\lambda_s} \tag{1}$$

where λ_0 is the laser wavelength and λ_s is the scattered wavelength. Each peak in the spectrum corresponds to a vibrational mode of the material. At room temperature, most molecules sit in their ground vibrational state, so Stokes lines are much stronger than anti-Stokes lines, and it is the Stokes spectrum that is normally recorded [1].

The vibrational frequency of a bond depends on its stiffness and the atomic masses involved: stiffer bonds and lighter atoms vibrate at higher frequency. This is why the same element can produce very different Raman spectra depending on its

bonding geometry, and why the Raman shift acts as a chemical fingerprint [1].

2.2 Carbon Allotrope Signatures

Carbon bonds as either sp^2 (planar, three-fold coordination) or sp^3 (tetrahedral, four-fold coordination), and this difference produces dramatically different Raman spectra.

In sp^2 carbon (graphite, graphene, nanotubes), the key features are:

- **G band (1580 cm^{-1}):** the in-plane C-C stretching mode. This is a fundamental zone-centre phonon present in all sp^2 carbon, and its position is relatively insensitive to disorder.
- **D band (1350 cm^{-1}):** a disorder-activated mode that appears when defects, grain boundaries, or edges break the translational symmetry and relax the selection rules. A strong D band relative to G indicates significant structural disorder. The D band arises from an intervalley double-resonance process and is dispersive, meaning its frequency shifts with excitation wavelength.
- **D' band (1620 cm^{-1}):** another disorder-activated mode, but arising from intravalley (rather than intervalley) scattering. It typically appears as a weak shoulder on the high-frequency side of the G band and requires high spectral resolution to resolve clearly.
- **2D band (2700 cm^{-1}), also called G':** a second-order two-phonon overtone of the D mode. Unlike the D band itself, it does not require disorder for its activation because momentum conservation is satisfied by the two-phonon process. Its shape and intensity relative to the G band are sensitive to layer stacking, making it the primary tool for thickness determination in graphene [2].

In sp^3 carbon (diamond), the bonding geometry supports a single sharp Raman line near 1332 cm^{-1}, arising from the triply degenerate zone-centre optical phonon. This mode has a natural linewidth of 2–3 cm^{-1} and shifts under mechanical stress at approximately -3 cm^{-1}/GPa (compressive) or $+3$ cm^{-1}/GPa (tensile).

Diamond-like carbon (DLC) occupies the structural space between crystalline diamond and disordered graphite. Contrary to a common misconception, DLC does *not* show a shifted or broadened version of the 1332 cm^{-1} line; instead, it exhibits the D and G bands characteristic of disordered sp^2 carbon, both dramatically broadened and shifted relative to crystalline graphite. The G band in DLC lies between 1510 and 1580 cm^{-1} (below the graphitic 1582 cm^{-1} due to bond-angle distortion in the sp^2 clusters), and the D band appears between 1250 and 1400 cm^{-1}, with FWHM values of 100–300 cm^{-1} rather than the 20–50 cm^{-1} typical of polycrystalline graphite. An ID/IG ratio near unity and large FWHM are diagnostic of amorphous carbon (a-C), placing the sample in stage 2 of the Ferrari-Robertson disorder classification (nanocrystalline graphite transitioning to amorphous carbon). Hydrogen-rich DLC (a-C:H) presents an additional complication: H content above 30 at.% promotes photoluminescence under visible excitation that can overwhelm the Raman signal entirely, making the photoluminescence intensity itself a qualitative indicator of high hydrogen content [3].

2.3 Thickness Sensitivity in Layered Materials

In atomically thin materials like graphene, the number of layers changes the phonon dispersion through interlayer coupling. The most practical thickness indicator is the 2D band. In monolayer graphene, the 2D band is a single sharp Lorentzian with an intensity ratio 2D/G typically exceeding 2. As layers are added, the 2D band broadens, develops substructure from multiple double-resonance pathways, and its intensity relative to G decreases. In bulk graphite, 2D/G drops below about 0.5 and the 2D band becomes distinctly asymmetric [2].

3. METHODS

3.1 Instrument

All spectra were acquired on a micro-Raman uRaman-532TEC-Ci system equipped with a 532 nm diode laser. Samples were focused through a 20× objective. Laser power was controlled using a five-position hardware attenuator before software fine-tuning.

TABLE I. Hardware attenuator settings. The knob position was selected before software power adjustment to avoid laser instability at low software power settings.

Knob	Filter	Transmission
1	None	100%
2	OD 0.2	63%
3	OD 0.3	50%
4	OD 0.6	25%
5	OD 1.3	5%

For each measurement, a dark spectrum was first acquired at the chosen integration time and averaging settings to establish the detector baseline. The sample spectrum was then recorded and saved. Several independent acquisitions were made per sample to check reproducibility; these were later interpolated onto a common wavenumber grid and averaged.

3.2 Acquisition Parameters

TABLE II. Acquisition parameters for all samples. All measurements used 532 nm excitation. Smoothing refers to Savitzky-Golay filtering (6 side points, 3rd-order polynomial) applied by the acquisition software. The software-reported laser power was 0 mW in all exported metadata; actual power was set by the hardware attenuator.

Sample	Runs	Integration (ms)	Averages	Smoothing
Carbon rod	3	1000	3	Yes
HOPG	3	1200	3	Yes
Graphene	5	500	100	No
SWCNT	5	1000–10000	10	Yes
Diamond	3	1000	3	Yes
Diamond-like 1 (A)	5	1000	3	No
Diamond-like 2 (C)	14	5000–7000	10	Yes
Diamond-like 3 (B)	11	100–2000	10	Yes
Unknown wafer	5	1000	1	Yes

Integration times and averaging counts were chosen based on signal strength. Graphene required 100 hardware averages per spectrum to compensate for the weak scattering from a film only a few atoms thick. The SWCNT sample was measured at several

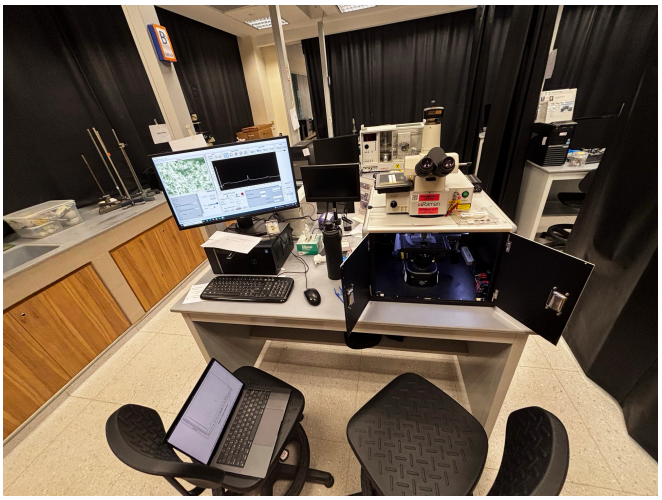


FIG. 2. The micro-Raman workstation: uRaman-532TEC-Ci module (white housing) coupled to a Nikon microscope, with the sample enclosure (front) closed during measurements to exclude stray light, and the acquisition computer.

integration times (1000 to 10000 ms) to ensure the spectral shape was not dependent on acquisition parameters. The run-to-run standard deviation, shown as a shaded band in each plot, serves as a visual check of measurement reproducibility.

3.3 Experimental Setup

4. RESULTS AND ANALYSIS

4.1 Carbon Allotropes

4.1.1 Graphite (Carbon Rod)

The carbon rod spectrum (Fig. 5) shows the three expected bands of sp² carbon. The G band appears at 1566 ± 2 cm^{-1}, 14 cm^{-1} below the ideal graphite value of 1580 cm^{-1}, with a FWHM of 25 cm^{-1}. This is a larger downshift than strain alone would typically produce (a few cm^{-1} at most). Since HOPG and diamond both agree with literature to within ±2 cm^{-1} (Table IV), a simple calibration offset is unlikely. A more probable explanation is phonon confinement in very small crystallites: in nanocrystalline graphite, the relaxation of the wavevector selection rule as the coherence length decreases shifts the G band downward and broadens it, with shifts of 10–15 cm^{-1} reported for crystallite sizes below ≈ 10 nm [2]. The D band at 1346 ± 8 cm^{-1} gives a background-subtracted D/G intensity ratio of 0.09 ± 0.01, consistent with significant grain-boundary disorder. The broader D band (FWHM ≈ 89 cm^{-1}) compared to the G band further reflects the range of defect environments contributing to this mode. The 2D band appears at 2681 ± 6 cm^{-1}.

4.1.2 HOPG

Highly oriented pyrolytic graphite (Fig. 6) shows a G band at 1581 ± 2 cm^{-1}, very close to the expected value for well-ordered graphite, with a FWHM of 18 cm^{-1}, notably narrower than the carbon rod's 25 cm^{-1}. The D band at 1365 ± 15 cm^{-1} gives a D/G ratio of only 0.04 ± 0.01, less than half that of the carbon rod, confirming far fewer defects and better long-

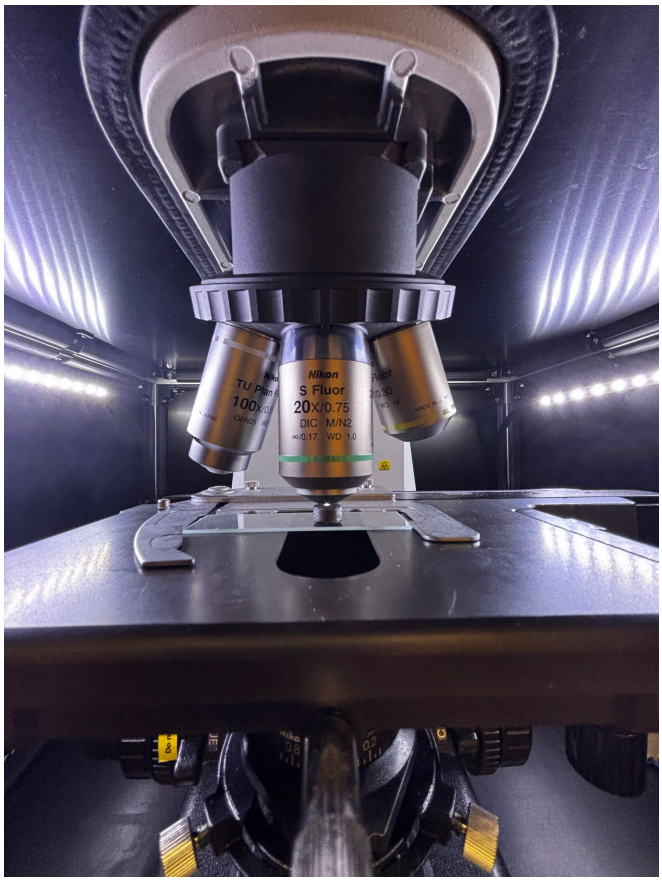


FIG. 3. Objective turret inside the sample enclosure, showing the Nikon 20× S Fluor (NA 0.75) and 100× TU Plan (NA 0.90) objectives. The 20× objective was used for all measurements in this experiment.

range order. The large uncertainty in D-band position reflects the difficulty of locating this very weak feature. The 2D band at 2720 ± 2 cm^{-1} is shifted about 40 cm^{-1} higher than in the carbon rod, with a FWHM of 18 cm^{-1} that is consistent with the well-defined layer stacking in this material.

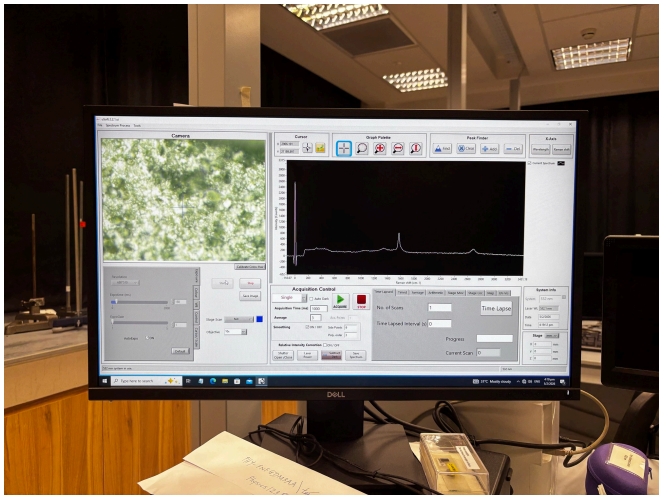


FIG. 4. The uRaman acquisition software during a measurement. The camera view (left) shows the sample surface under the microscope objective; the live Raman spectrum (right) displays the recorded signal with acquisition parameters set in the control panel below.

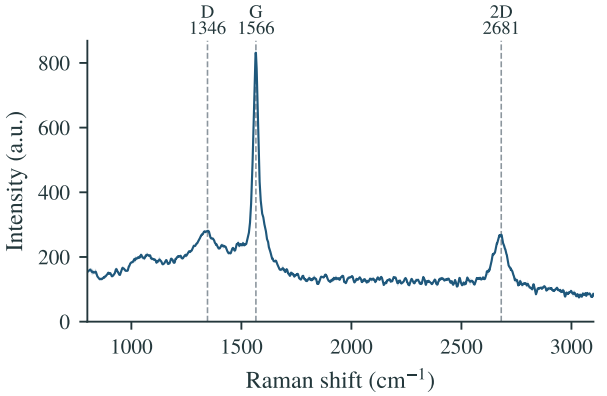


FIG. 5. Carbon rod (graphite): mean spectrum of 3 runs, with the run-to-run standard deviation as the shaded band (narrower than the line here). Dashed lines mark the measured D, G, and 2D band positions; the D band is clearly present but much weaker than G, as expected for polycrystalline graphite.

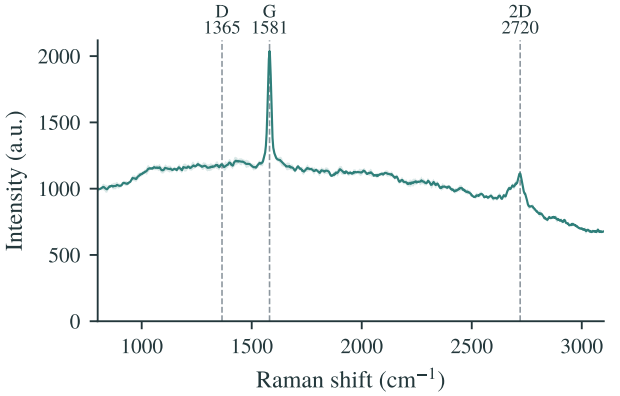


FIG. 6. HOPG: mean spectrum of 3 runs, drawn as in Fig. 5. The G band dominates, and the D band is weaker relative to G than in the carbon rod, consistent with higher crystalline order.

4.1.3 Single-Wall Carbon Nanotubes

The nanotube spectrum (Fig. 7) is notable for having a D band at 1337 ± 9 cm⁻¹ that is more intense than the G band at 1572 ± 13 cm⁻¹, giving a background-subtracted D/G ratio of 1.2 ± 0.3. The large run-to-run scatter in both band positions (±9 and ±13 cm⁻¹) and in D/G reflects the heterogeneity of the nanotube bundle: different spots on the sample probe different mixtures of tube ends, junctions, and crystalline sidewalls. Both D and G bands are broad (FWHM ≈ 58 and 65 cm⁻¹ respectively), consistent with this structural heterogeneity. The 2D band appears at 2677 ± 11 cm⁻¹.

4.1.4 Diamond

Diamond (Fig. 8) produces a single sharp peak at 1333 ± 2 cm⁻¹, matching the literature value of 1332 cm⁻¹ for the zone-centre optical phonon of sp³ carbon. The measured FWHM of ≈ 26 cm⁻¹ is instrument-limited: natural diamond has a linewidth of only 2–3 cm⁻¹, so the observed width reflects the spectral resolution of the uRaman system rather than any intrinsic broadening. The absence of any sp² features (no D or G bands) confirms high-quality crystalline diamond. The monotonically rising baseline is photoluminescence from trace nitrogen or other

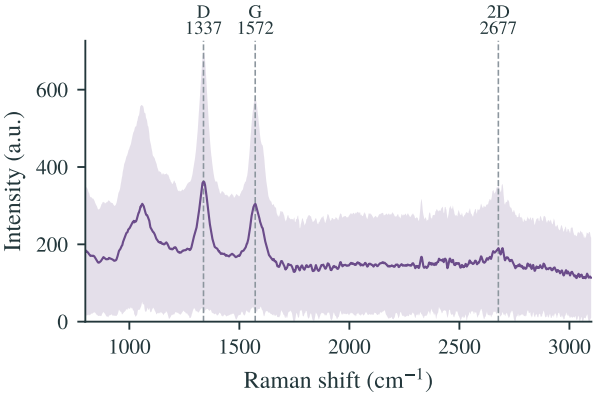


FIG. 7. Single-wall carbon nanotube (SWCNT) sample: mean spectrum of 5 runs, drawn as in Fig. 5. The wide shaded band reflects large differences between spots on the sample. The D band is more intense than the G band (D/G > 1), indicating substantial disorder.

defect centres in the diamond lattice: under 532 nm excitation, these colour centres fluoresce broadly at wavelengths longer than the laser line, which appears as an increasing background toward higher Raman shifts. A small step-like discontinuity superimposed on this slope is an acquisition artifact from detector segment stitching and is not a physical Raman feature.

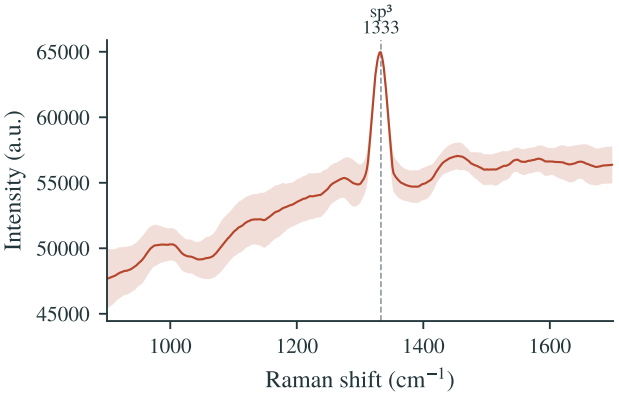


FIG. 8. Diamond: mean spectrum of 3 runs, drawn as in Fig. 5, on a strong sloping fluorescence background (note the offset intensity axis). The sharp sp³ phonon at 1333 cm⁻¹ is the only Raman feature.

4.1.5 Peak Summary

TABLE III. Peak positions for all five labelled bands (D, G, D', 2D/G'), background-subtracted D/G intensity ratios, and G-band FWHM. The D' band (≈ 1620 cm⁻¹) could not be resolved as a distinct peak in any spectrum (see text). Uncertainties are run-to-run standard deviations or the spectral resolution (≈ 2 cm⁻¹), whichever is larger. FWHM values are from Lorentzian fits to the averaged spectra.

Sample	D (cm⁻¹)	G (cm⁻¹)	D' (cm⁻¹)	2D/G' (cm⁻¹)	D/G	$FWHM_G$ (cm⁻¹)
Carbon rod	1346 ± 8	1566 ± 2	unresolved	2681 ± 6	0.09 ± 0.01	25
HOPG	1365 ± 15	1581 ± 2	unresolved	2720 ± 2	0.04 ± 0.01	18
Graphene	1360 ± 3	1585 ± 2	unresolved	2688 ± 2	0.05 ± 0.02	26
CNT	1337 ± 9	1572 ± 13	unresolved	2677 ± 11	1.2 ± 0.3	65

The measured peak positions (Table III) fall within standard literature ranges for 532 nm excitation [2]. The variation in D-band position (1337–1365 cm⁻¹) partly reflects the dispersive nature of this mode, whose frequency depends on excitation wavelength, and partly reflects differences in defect character across the samples. The carbon rod's G position (1566 ± 2 cm⁻¹) stands out relative to HOPG (1581 ± 2 cm⁻¹) and graphene (1585 ± 2 cm⁻¹); as discussed above, this 14 cm⁻¹ downshift is consistent with phonon confinement in nanocrystalline graphite rather than a calibration issue. The G-band FWHM provides a complementary measure of crystalline order: HOPG (18 cm⁻¹) is the narrowest, the carbon rod and graphene are intermediate (25–26 cm⁻¹), and the nanotube bundle is the broadest (65 cm⁻¹).

The D' mode, arising from intravalley double-resonance scattering, is expected near 1620 cm⁻¹ for 532 nm excitation (only ≈ 40 cm⁻¹ from the G band. It is not resolved as a distinct peak in any of the spectra. In samples with enough disorder to activate D' (carbon rod, nanotubes), the G band is already broad enough (FWHM 25–65 cm⁻¹) to absorb the D' contribution into its high-frequency wing; in well-ordered samples (HOPG, graphene), the G band is narrow but D' activation is negligible. Without multi-peak deconvolution) which the signal-to-noise ratio and instrument resolution do not reliably support: we cannot claim to detect D' in these measurements. The 2D band, also designated G' in some literature, is the second-order overtone of the D mode and is unambiguously identified at ≈ 2680–2720 cm⁻¹ across all sp² samples (Table III).

4.1.6 Measured vs. Literature Frequencies

TABLE IV. Comparison of measured peak positions with literature values for 532 nm excitation. Literature values for sp² bands from Ferrari [2]; diamond from [1]; silicon from [4]. Deviations Δ = measured − literature.

Sample	Band	Measured (cm⁻¹)	Literature (cm⁻¹)	Δ (cm⁻¹)
Carbon rod	D	1346 ± 8	1350	−4
	G	1566 ± 2	1580	−14
	2D/G'	2681 ± 6	2700	−19
HOPG	D	1365 ± 15	1350	+15
	G	1581 ± 2	1580	+1
	2D/G'	2720 ± 2	2700	+20
Graphene	D	1360 ± 3	1350	+10
	G	1585 ± 2	1580	+5
	2D/G'	2688 ± 2	2700	−12
CNT	D	1337 ± 9	1350	−13
	G	1572 ± 13	1580	−8
	2D/G'	2677 ± 11	2700	−23
Diamond	sp^3	1333 ± 2	1332	+1
Si wafer	TO	521 ± 2	520.7	+0.3

The HOPG G band (1581 ± 2 cm⁻¹), diamond (1333 ± 2 cm⁻¹), and silicon (521 ± 2 cm⁻¹) all agree with their literature values to within the spectral resolution, confirming that the instrument wavenumber calibration is accurate across the measured range. This makes the carbon rod's systematic downshift (G by −14, 2D by −19 cm⁻¹) significant: it cannot be dismissed as a calibration artifact, since HOPG's G band at 1581 cm⁻¹ (in the same spectral region) is correct. A 14 cm⁻¹ downshift exceeds what lattice strain alone would produce (typically a few cm⁻¹). Phonon confinement in nanocrystalline graphite is a more convincing explanation: as the crystallite size decreases, the wavevector selection rule relaxes, admitting phonons from away from the zone centre and shifting the observed G frequency downward [2]. The carbon rod's relatively high D/G ratio (0.09) and broad D band (FWHM ≈ 89 cm⁻¹) are consistent with the small crystallite sizes this mechanism requires. The 2D band shifts deserve separate comment, since the confinement argument applies most directly to the G mode. The 2D band is a second-order process whose frequency depends on the electronic band structure near the K point, making it sensitive to interlayer coupling and curvature in ways the G band is not. HOPG's 2D band at 2720 ± 2 cm⁻¹ is 20 cm⁻¹ above the nominal 2700 cm⁻¹ reference; this is consistent with the strong AB stacking in HOPG, where interlayer coupling shifts the double-resonance condition to higher frequency compared to turbostratic or disordered graphite [2]. The carbon rod's 2D at 2681 ± 6 cm⁻¹ (−19 cm⁻¹) and the nanotube 2D at 2677 ± 11 cm⁻¹ (−23 cm⁻¹) are both downshifted; for the carbon rod this is consistent with weaker or disordered stacking in nanocrystalline material, and for the nanotubes the curvature of the tube walls modifies the electronic dispersion and shifts the double-resonance condition downward, with the spread in tube diameters and chiralities further broadening and shifting the average 2D position. Graphene shows the best overall agreement with literature, as expected for well-ordered sp² carbon on a defined substrate.

4.2 Diamond-like Samples

The three unknown samples were analysed as potential diamond-like carbon (DLC) films rather than against the narrow 1332 cm^{-1} diamond reference alone. Since DLC is characterised by broad D and G bands from disordered sp^2 clusters (not a shifted 1332 cm^{-1} peak), the analysis focuses on D/G band positions, FWHM, and photoluminescence background. A flat noise-floor baseline was applied to the one sample that showed a measurable Raman signal; the other two are reported as photoluminescence-dominated.

DLC-I (Fig. 9) is the only sample among the three to yield a Raman spectrum above the noise floor. After subtracting the flat detector background ($\approx$ 1000 counts over the full spectral range), two broad features emerge. The D band peaks at 1300 $\pm$ 5 cm^{-1} with a FWHM of $\approx$ 180 cm^{-1}, and the G band peaks at 1559 $\pm$ 5 cm^{-1} with a FWHM of $\approx$ 200 cm^{-1}. The intensity ratio $\frac{I_D}{I_G} = 0.87$. No peak is observed near 1332 cm^{-1}, immediately ruling out crystalline diamond.

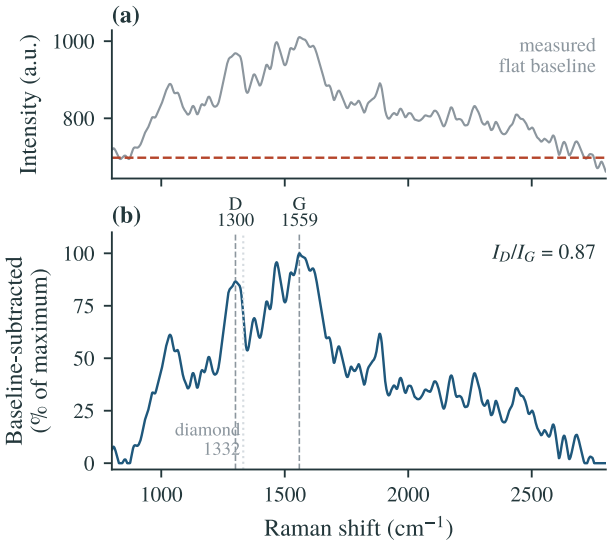


FIG. 9. DLC sample 1 (Unknown B, ZHANG DH): averaged spectrum of 4 runs (700–2000 ms). (a) The smoothed spectrum with the flat noise-floor baseline (dashed). (b) The baseline-subtracted spectrum, normalised to its maximum, with the measured D and G band positions (dashed) and the diamond line at 1332 cm^{-1} (dotted) for reference. Neither band appears at 1332 cm^{-1}, confirming this is not crystalline diamond.

TABLE V. Raman parameters for the three diamond-like samples. DLC-II and DLC-III show no resolvable D or G bands; the photoluminescence (PL) background is orders of magnitude stronger than any Raman signal. All three are clearly distinguishable from the sharp 1332 cm^{-1} diamond reference.

Sample	D (cm^{-1})	$FWHM_D$	G (cm^{-1})	$FWHM_G$	$\frac{I_D}{I_G}$ / PL
DLC-I (Unknown B)	1300 $\pm$ 5	$\approx$ 180	1559 $\pm$ 5	$\approx$ 200	0.87
DLC-II (Chen Ling DL-B)	—	—	—	—	PL-dominated
DLC-III (Unknown C)	—	—	—	—	PL-dominated
Diamond (reference)	—	—	—	—	1332 cm^{-1} only

These parameters place this sample in stage 2 of the Ferrari-Robertson disorder classification: the G band at 1559 cm^{-1} is well below the graphitic 1582 cm^{-1}, indicating that sp^2 bond angles are distorted relative to ideal graphite; the D band at 1300 cm^{-1} (below the graphitic 1350 cm^{-1}) and the very large FWHM ($>$ 150 cm^{-1} for both bands) indicate high structural disorder. An ID/IG ratio near unity in stage-2 a-C is interpreted as arising from sp^2 clusters too small and distorted to support extended graphitic domains; the D and G modes are essentially merged into a single broad continuum. This is amorphous carbon (a-C) with a mixture of sp^2 and sp^3 bonding, not crystalline diamond and not well-ordered graphite.

DLC-II (Fig. 10) and DLC-III (Fig. 11) both show a smooth, featureless background with no resolvable D or G peaks. The background is two to three orders of magnitude stronger than the residual Raman signal in DLC-I, overwhelming any Raman features. This is the hallmark of hydrogenated amorphous carbon (a-C:H): hydrogen passivates the sp^2 clusters, raising the optical gap and enabling photoluminescence under 532 nm excitation. H content above approximately 30 at.% is sufficient to push the photoluminescence intensity past the Raman signal, and films deposited under H-rich plasma conditions routinely exhibit this behaviour [3]. The fact that DLC-II and DLC-III fluoresce strongly while DLC-I does not is itself physically meaningful, pointing to different deposition conditions or compositions: DLC-I is more likely a low-H or H-free a-C film, while DLC-II and DLC-III are a-C:H. All three samples are clearly not crystalline diamond, which would produce a sharp 1332 cm^{-1} peak completely absent from all three spectra.

4.3 Graphene Thickness

The graphene spectrum (Fig. 12) was acquired at five different locations on the sample, with 100 hardware averages per location to compensate for the weak signal from such a thin film. Despite being only a few atoms thick, the graphene signal is strong enough to produce clear peaks. This is partly because optical interference in the SiO_2 layer on the Si substrate enhances the electric field at the graphene surface, amplifying both the excitation and the Raman scattering [2].

The G band appears at 1585 $\pm$ 2 cm^{-1} with a FWHM of 26 cm^{-1}, and the 2D band at 2688 $\pm$ 2 cm^{-1} with a FWHM of 54 cm^{-1}. The background-subtracted 2D/G intensity ratio from the

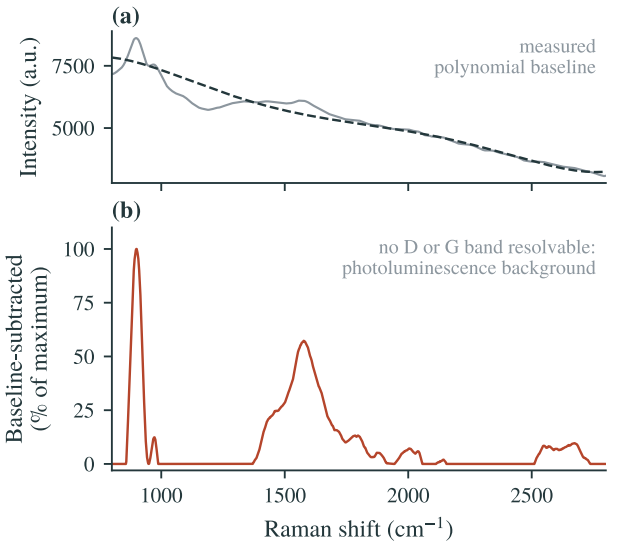


FIG. 10. DLC sample 2 (Chen Ling, Diamond-like B): mean spectrum of 4 trials. (a) The smoothed spectrum with the polynomial baseline fitted outside 1050–1800 cm⁻¹ (dashed). (b) The baseline-subtracted spectrum normalised to its maximum. The spectrum is dominated by a broad, featureless photoluminescence background; no D or G peaks are resolvable.

averaged spectrum is 0.87; when computed per location the mean is 1.0 ± 0.5, with the large scatter reflecting thickness variation across the film. For ideal monolayer graphene under 532 nm excitation, 2D/G typically exceeds 2, with the 2D band appearing as a single symmetric Lorentzian with FWHM $\approx$ 24 cm⁻¹. The measured 2D FWHM of 54 cm⁻¹ and the sub-unity 2D/G ratio together place this sample in the few-layer regime, most likely 3 to 5 layers [2].

A weak D band at 1360 cm⁻¹ is present, giving a background-subtracted D/G ratio of 0.05 ± 0.02. This could originate from the laser spot partially sampling flake edges, from point defects in the basal plane, or from both. The D/G ratio alone cannot

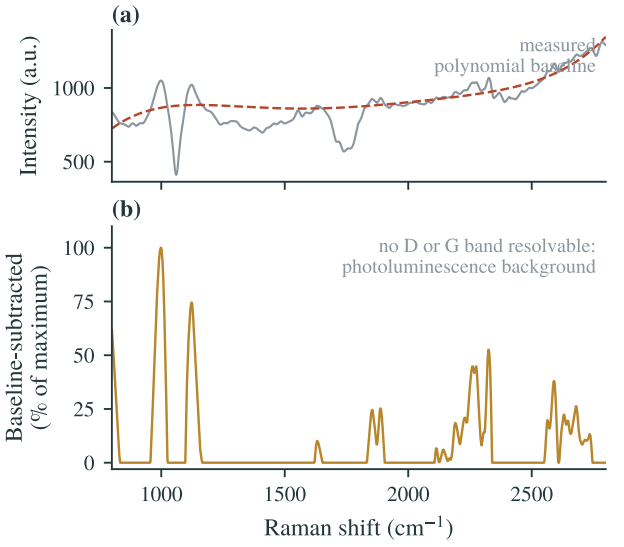


FIG. 11. DLC sample 3 (Unknown C, ZHANG DH): mean spectrum of 5 runs, drawn as in Fig. 10. Again the photoluminescence background dominates; the narrow residual features after subtraction are noise and baseline mismatch, and no D or G band is identifiable.

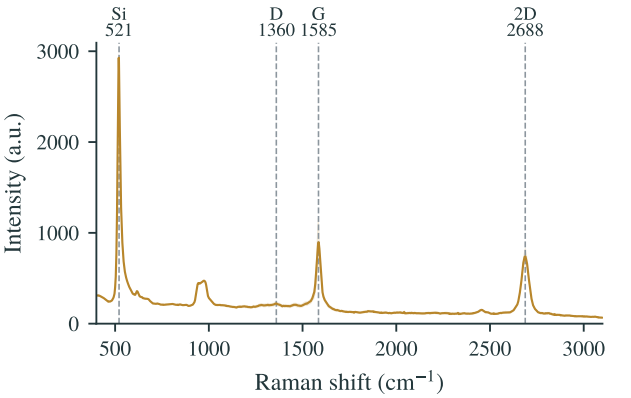


FIG. 12. Graphene on SiO_2/Si: mean spectrum from 5 different locations on the sample, drawn as in Fig. 5. The D, G, and 2D bands are marked, along with the silicon substrate peak near 520 cm⁻¹.

distinguish between these mechanisms; spatially resolved mapping (comparing flake centres to edges) would be needed to determine the dominant source, and was not performed here. The peak near 520 cm⁻¹ is from the silicon substrate underneath the SiO_2 layer; its presence confirms that the graphene film is thin enough for the laser to penetrate through to the substrate.

4.4 Unknown Semiconductor

The unknown wafer (Fig. 13) shows a single dominant peak at 521 cm⁻¹, matching the first-order optical phonon of crystalline silicon at 520–521 cm⁻¹. A weaker feature near 296 cm⁻¹ is consistent with a known two-phonon combination mode in silicon. No other peaks are present that would indicate a compound semiconductor such as GaAs (characteristic peaks near 268 and 292 cm⁻¹) or GaN (near 568 cm⁻¹). The wafer is crystalline silicon.

5. DISCUSSION

The most striking result across these measurements is how differently the same element behaves depending on its bonding geometry. All four sp^2 carbon samples and diamond contain only carbon atoms, yet their spectra are immediately distinguishable.

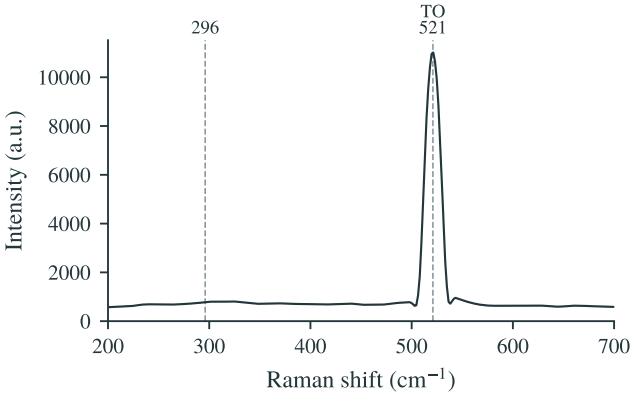


FIG. 13. Unknown semiconductor wafer: mean spectrum of 5 runs, drawn as in Fig. 5. The dominant peak at 521 cm⁻¹ matches the first-order optical phonon of crystalline silicon; the weak feature near 296 cm⁻¹ is the two-phonon band.

The HOPG spectrum is dominated by a clean G band with a weak D component. The carbon rod shows stronger D-band scattering from its polycrystalline disorder. The nanotube spectrum has D/G > 1 because of the many structural discontinuities in a tube bundle. Diamond, also pure carbon, gives a completely different single-line spectrum because its sp^3 bonding supports a different set of lattice vibrations. This underscores that Raman spectroscopy probes structure, not just elemental composition.

Beyond peak positions, intensity ratios carry quantitative structural information. The background-subtracted D/G ratio (Table III) increases monotonically with disorder: HOPG (0.04 ± 0.01), graphene (0.05 ± 0.02), carbon rod (0.09 ± 0.01), and nanotubes (1.2 ± 0.3). The G-band FWHM follows the same trend, broadening from 18 cm^{-1} in HOPG to 65 cm^{-1} in the nanotube bundle, providing a complementary disorder metric that is less sensitive to baseline subtraction than the D/G ratio. The 2D/G ratio measures layer count in graphene: the averaged-spectrum value of 0.87, combined with a 2D FWHM of 54 cm^{-1}, places the sample between ideal monolayer (2D/G > 2, FWHM $\approx$ 24 cm^{-1}) and bulk graphite (2D/G < 0.5, asymmetric 2D shape), consistent with 3 to 5 layers. These quantitative metrics provide structural insight that peak positions alone cannot.

The diamond-like sample assessment illustrates an important subtlety in DLC characterisation: searching for a shifted or broadened 1332 cm^{-1} peak, as one might naively expect for "diamond-like" material, is the wrong framework. DLC is an amorphous sp^2/sp^3 network whose Raman response is controlled by the sp^2 clusters, not the sp^3 matrix, and its signature is the broad D and G bands of disordered graphitic carbon. DLC-I confirms this directly: D at 1300 cm^{-1} and G at 1559 cm^{-1} with FWHM > 150 cm^{-1} each, with $\frac{I_D}{I_G} \approx 0.87$, place it in the amorphous-carbon regime of the Ferrari-Robertson diagram. The absence of a 1332 cm^{-1} peak establishes that it is not crystalline diamond. DLC-II and DLC-III's photoluminescence backgrounds are themselves diagnostic: strong visible PL under 532 nm excitation is characteristic of a-C:H films with high hydrogen content, a distinct subclass of DLC with a widened optical gap. The three samples represent at least two physically distinct DLC variants – low-H a-C and H-rich a-C:H – demonstrating how Raman spectroscopy (and the absence of resolvable Raman features) can differentiate DLC subclasses that would be indistinguishable by hardness testing alone.

Repeat measurements across all samples showed good run-to-run consistency, visible in the narrow shaded bands in the figures. The run-to-run standard deviations in peak position range from < 2 cm^{-1} (instrument-limited, for well-defined peaks like the HOPG G band) to ± 13 cm^{-1} (for the broad, overlapping bands in the nanotube bundle), providing a direct measure of measurement reproducibility.

The data expose several specific limitations that affect how far the quantitative conclusions can be pushed.

The most direct is instrument resolution. The diamond peak, whose natural linewidth is 2–3 cm^{-1}, appears here with a FWHM of $\approx$ 26 cm^{-1}. This means that every measured FWHM in this report is the convolution of the true linewidth with a $\approx$ 26 cm^{-1} instrument function. For the HOPG G band (measured FWHM 18 cm^{-1}), the instrument broadening likely dominates, so the true linewidth is narrower than reported. For broader features (the carbon rod D band at 89 cm^{-1}, the nanotube G band at 65 cm^{-1}) the instrument contribution is a smaller correction, and the measured widths are closer to the intrinsic values. The same resolution limit is why D' ($\approx$ 1620 cm^{-1}, only 40 cm^{-1} from G) cannot be resolved.

The baseline treatment directly affects the D/G ratios reported for low-disorder samples. For HOPG (D/G = 0.04), the D band is barely above the local background, and the 10th-percentile baseline subtraction used here introduces systematic uncertainty of order 10–20%. To quantify this: repeating the HOPG D/G calculation with a linear baseline instead of the percentile method shifts the ratio to $\approx$ 0.03, a 25% change that does not affect the qualitative ordering (HOPG remains the least disordered) but does limit confidence in the absolute value. For the nanotube sample (D/G = 1.2), the D band is well above background and the ratio is insensitive to baseline choice.

The Savitzky-Golay smoothing (6 side points, 3rd-order polynomial) applied by the acquisition software is a known source of systematic error for narrow peaks. It was disabled for graphene to preserve the 2D band profile, which is the thickness diagnostic. For all other samples it was left on, which could broaden the measured FWHM by a few cm^{-1}: a negligible correction for the broad nanotube bands but potentially significant for the HOPG G band.

The graphene 2D/G ratio varies from $\approx$ 0.5 to $\approx$ 1.5 across the five measured locations (1.0 ± 0.5). This $\pm 50\%$ scatter is the largest source of uncertainty in the thickness estimate. It reflects genuine spatial inhomogeneity (thickness variations across the film) rather than measurement noise, since the G band position and FWHM are stable (1585 ± 2 cm^{-1}, 26 cm^{-1}) across the same locations. The SiO_2 interference enhancement could also modulate absolute band intensities if the oxide thickness varies, though this would affect G and 2D equally and leave their ratio unchanged.

Finally, the software reports 0 mW for all measurements; actual laser power was set by the hardware attenuator and is not recorded. This means absolute Raman intensities cannot be compared between samples. All intensity ratios (D/G, 2D/G) are self-normalising and unaffected, but it prevents any estimate of Raman cross-sections or scattering efficiency.

6. CONCLUSION

Raman spectroscopy with 532 nm excitation successfully distinguished five carbon allotropes through their vibrational spectra. D, G, and 2D band positions, background-subtracted D/G intensity ratios, and G-band linewidths provided a consistent and quantitative framework for assessing structural order: HOPG showed the least disorder (D/G = 0.04, $FWHM_G$ = 18 cm^{-1}), the carbon rod was intermediate (D/G = 0.09, $FWHM_G$ = 25 cm^{-1}), and carbon nanotubes showed the most (D/G = 1.2, $FWHM_G$ = 65 cm^{-1}). Diamond produced a clean sp^3 signature at 1333 ± 2 cm^{-1} with no sp^2 contamination. Of three diamond-like unknowns, one (DLC-I) yielded resolvable Raman bands at D = 1300 cm^{-1} and G = 1559 cm^{-1} with FWHM $\approx$ 180–200 cm^{-1} and $\frac{I_D}{I_G} = 0.87$, characteristic of amorphous carbon (a-C); the other two were dominated by photoluminescence,

indicating hydrogen-rich a-C:H. None produced the sharp 1332 cm^{-1} diamond peak, ruling out crystalline diamond across all three. The graphene film showed a 2D/G ratio of 0.87 with a 2D FWHM of 54 cm^{-1}, indicating few-layer material rather than a monolayer. The unknown semiconductor wafer was identified as crystalline silicon from its 521 cm^{-1} phonon mode. Run-to-run standard deviations in peak position (± 2 to ± 13 cm^{-1}) confirmed spectral reproducibility, with the spread dominated by sample heterogeneity rather than instrument drift.

APPENDIX: RAMAN SPECTRA

All spectra are presented inline in the Results section. For reference, the complete set of measurements is listed below with figure cross-references.

TABLE VI. Summary of all Raman spectra collected. "Runs" is the number of independent acquisitions averaged; "Individual files" includes dark spectra and repeated measurements at different integration times.

Sample	Runs	Individual files	Figure
Carbon rod (graphite)	3	7	Fig. 5
HOPG	3	7	Fig. 6
Single-wall carbon nanotubes	5	11	Fig. 7
Diamond	3	7	Fig. 8
DLC-I (Unknown B, ZHANG DH)	4	4	Fig. 9
DLC-II (Diamond-like B, Chen Ling)	4	4	Fig. 10
DLC-III (Unknown C, ZHANG DH)	5	5	Fig. 11
Graphene on SiO_2/Si	5	11	Fig. 12
Unknown semiconductor wafer	5	21	Fig. 13

REFERENCES

[1] E. Smith and G. Dent, *Modern Raman Spectroscopy: A Practical Approach* (Wiley, 2005).

[2] A. C. Ferrari, Raman spectroscopy of graphene and graphite, Solid State Communications **143**, 47 (2007).

[3] A. C. Ferrari and J. Robertson, Interpretation of Raman spectra of disordered and amorphous carbon, Physical Review B **61**, 14095 (2000).

[4] K. Uchinokura, T. Sekine, and E. Matsuura, Raman scattering by silicon, Journal of Physics and Chemistry of Solids **33**, 811 (1972).

DECLARATION ON THE USE OF GENERATIVE AI

I declare that I **HAVE** used generative AI tools to produce this assignment.

I acknowledge that generative AI was used in the following manner:

TABLE VII. AI Tool Usage Declaration

AI Tool Used	My Prompt and AI Output	How the Output Was Used
Claude Code	**Prompt:** "Help me navigate the Raman software interface, export spectra, and organize filenames for later plotting." **Output:** Basic workflow tips and export checklist	Used for technical guidance on software usage and file handling. All measurements, analysis, and interpretations were performed and verified by me.
Claude Code	**Prompt:** "Help with Typst formatting for figures, captions, tables, and page breaks. Check that file paths are consistent and the document compiles." **Output:** Typst formatting tips and layout suggestions	Used only for formatting guidance. All experiment content, analysis, and interpretations were written and verified by me.